

NSF Workshop on Security Challenges and Research Directions for Space and Non-Terrestrial Networks

Motivation

Over the past two decades, *non-terrestrial networks* (NTNs) have evolved into foundational infrastructure for both defense and civilian ecosystems [1–3]. They enable global connectivity, positioning, navigation, and timing (PNT), and provide operational support to critical lifeline systems, including electric power grids, water distribution, and transportation networks. In many regions, NTNs serve as the only viable means of resilient backhaul and wide-area coverage, reinforcing their strategic importance [4]. Emerging space-based networks may eventually host distributed data centers within satellite constellations, enabling new architectures for globally accessible cloud services [5, 6].

At the same time, the wireless nature of these systems, which operate over shared and often contested spectrum, exposes them to persistent and increasingly sophisticated threats. As adversarial capabilities have advanced, electronic-warfare techniques once confined to military domains are now being adapted for use against civilian communications infrastructure. NTNs are especially vulnerable: their reliance on long-range radio links, broad signal footprints, and limited prospects for physical hardening or rapid maintenance significantly enlarges the attack surface. Moreover, the tight coupling between space-based assets (e.g., satellites and/or high-altitude platforms) and terrestrial infrastructure introduces unique cross-domain dependencies, creating distinct security challenges that span the space, air, and ground segments of the network.

An example NTN architecture (see Figure 1) consists of LEO satellites interconnected via inter-satellite links, feeder links connecting satellites to the terrestrial core, and access links connecting satellites to user terminals, such as Very Small Aperture Terminals (VSATs). Hence, the security challenges of NTNs stem from such a large attack surface, and the tight coupling of links across all space, air, and ground segments. By integrating space-based assets with terrestrial and airborne infrastructure over shared wireless media, NTNs expose multiple interfaces, including access layer, feeder, and inter-satellite links, to potential exploitation. As a result, vulnerabilities at any single point can be leveraged to disrupt, intercept, or manipulate network operations across the entire system, making end-to-end security particularly difficult to guarantee.

At the access layer, user terminals operate over wide broadcast footprints, making them susceptible to smart jamming, spoofing, and impersonation attacks, among other techniques. Rogue base stations or malicious transmitters can masquerade as legitimate network elements, misleading user equipment or injecting false control information. Indeed, because signals can travel long distances over open spectrum, attackers do not need physical proximity to exploit these links. Vulnerabilities at any interface (inter-satellite, feeder, or access) can be exploited to mount attacks influenced by electronic warfare, enabling the disruption and interception of both satellite and terrestrial network operations. Furthermore, a compromised or rogue satellite can inject malicious signals or disrupt inter-satellite links, propagating faults across the constellation. Feeder links between satellites and the ground core represent particularly sensitive choke points: attacks there can affect large geographic regions and directly impact the integrity of the terrestrial core.

Unfortunately, current NTN systems lack security mechanisms specifically designed for their unique threat environment, instead relying largely on standard encryption techniques, which are insufficient given the severity and sophistication of electronic warfare and spoofing threats (as well as the pace of emerging quantum methods). This gap underscores the need for a focused research roadmap that identifies the

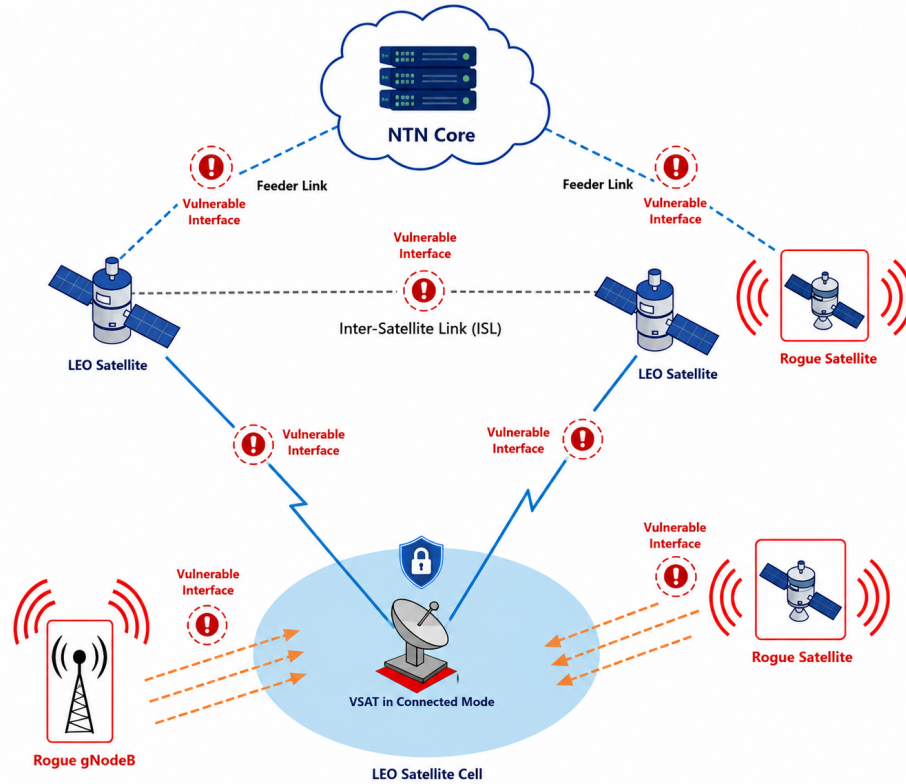


Figure 1: Representative security threat surface for space and NTN networks.

key security challenges and develops principled solutions to mitigate these vulnerabilities. The timing is particularly critical, given that as industry moves toward rapid deployment of NTNs, the opportunity to integrate robust security mechanisms into system architectures and standards is rapidly narrowing, and delayed action risks locking in avoidable weaknesses for the foreseeable future.

Figure 1 illustrates a representative security threat model for an example future NTN architecture composed of LEO satellites, terrestrial user terminals, and an NTN core network. The figure highlights multiple attack surfaces across the end-to-end communication path, emphasizing the need for a holistic approach to NTN security.

Although a growing body of research has examined individual aspects of NTN security (see e.g., [7, 8]), there has been no comprehensive effort to systematically assess the full threat landscape, identify critical research gaps, and develop a coordinated research roadmap for securing the future of NTN ecosystems in the United States.

Goals

The workshop will bring together established research and industry leaders, junior faculty, and postdoctoral researchers to foster cross-pollination of ideas and develop new approaches to securing future NTN networks.

Discussions will focus on the following threat areas and research challenges:

1. **Air Interface Vulnerabilities and Signal-Based Attacks:** jamming, spoofing, protocol exploitation, and intelligent adversarial interference.
2. **Rogue Satellites and Space-Based Threats:** malicious satellites, unauthorized transmissions, orbital interference, and satellite impersonation.
3. **Physical and Infrastructure Attacks:** attacks on ground stations, feeder links, terrestrial infrastructure, radio-access networks, and supply-chain vulnerabilities.
4. **Core Network and Cloud Security:** vulnerabilities arising from virtualization, cloud-native architectures, and satellite-enabled data services.
5. **AI- and Quantum-Enabled Threats:** adaptive and autonomous attacks powered by artificial intelligence, including intelligent jamming, protocol manipulation, adversarial machine learning, and automated vulnerability discovery, as well as quantum-computing threats to cryptographic protocols, authentication mechanisms, and secure key management.
6. **Quantum Technologies for Secure NTN Operations:** opportunities and risks associated with quantum communications, quantum sensing, and resilient PNT systems.
7. **Data Security, Privacy, Trust, and Confidence:** protection of user data, identity management, cross-domain information sharing, and privacy-preserving NTN services.

The workshop will explore the policy, regulatory, and standards implications of NTN security, including the transition to post-quantum cryptographic mechanisms, the standardization of cellular access protocols, and the development of governance frameworks that support secure, interoperable, and resilient non-terrestrial communication systems.

The research agenda produced by the workshop is expected to inform stakeholders across government, academia, and industry and to contribute to the development of a national strategy to secure current and future NTNs in the United States. Over the longer term, the workshop will catalyze cross-disciplinary research teams and raise awareness of fundamental research challenges associated with securing future generations of NTNs.

Relevant NSF-Sponsored Workshops

Several NSF-sponsored workshops conducted in recent years have addressed research challenges closely related to NTNs, including wireless communications, cybersecurity, network resilience, satellite communications, and next-generation networking systems.

- **NSF Workshop on Open and Integrated Network Testbeds for Space-Air-Ground Networks (2024).** This event focused on the development of open and integrated testbeds spanning satellite, aerial, and terrestrial communication systems. The workshop produced recommendations and a research roadmap for future experimentation platforms supporting next-generation wireless networks.
- **Trusted CI / NSF Cybersecurity Summit (2025).** This workshop focused on cybersecurity challenges facing the national research cyberinfrastructure ecosystem. Topics included operational security, trust management, cyber resilience, and best practices for securing research infrastructure.

Although several NSF-sponsored workshops have examined satellite communications, space-air-ground integrated networking, wireless security, and next-generation communication systems, the security of NTN has not been a primary focus of these efforts. This gap is becoming increasingly significant as NTNs evolve from specialized satellite systems into an integral component of 5G-Advanced and future 6G architectures.

Organizing Committee

The workshop will be organized in collaboration with leading experts in network and systems security, including academic members of the Next G Alliance Research Council, who are internationally recognized researchers and influential industry leaders. Their collective expertise will strengthen the workshop across five critical dimensions: technical vision and framing, scientific credibility, industry engagement, standards and policy relevance, and the translation of research outcomes into actionable initiatives. By bridging academia, industry, and standards communities, these leaders will help ensure that the workshop produces both rigorous scientific insights and a practical roadmap for advancing secure next-generation communication systems.

The tentative list of organizers is as follows:

- **Dr. Alex Sprintson (George Mason University), Organizing Committee Chair.** Dr. Alex Sprintson is Chair and Professor in the Department of Electrical and Computer Engineering at George Mason University. His research spans wireless communications, network security, coding theory, privacy-preserving computation, and next-generation communication systems, with significant contributions to NSF programs, telecommunications research, and standards-related initiatives. He previously served as a Program Director at the National Science Foundation and held research leadership positions in both academia and industry.
- **Dr. Guevara Noubir (Northeastern University).** Dr. Guevara Noubir is a Professor of Electrical and Computer Engineering at Northeastern University and a leading researcher in wireless security, network security, and privacy. His work has uncovered fundamental vulnerabilities in wireless systems and influenced both security research and practical defenses across cellular, Wi-Fi, and cyber-physical systems.
- **Dr. Aanjhan Ranganathan (Northeastern University).** Dr. Aanjhan Ranganathan is an Associate Professor in the Khoury College of Computer Sciences at Northeastern University. His research focuses on mobile systems, wireless security, sensing, and applied machine learning, with notable contributions in side-channel analysis, mobile device security, RF sensing, and emerging wireless technologies.
- **Dr. Yongdae Kim (KAIST).** Dr. Yongdae Kim is a Professor in the School of Electrical Engineering at the Korea Advanced Institute of Science and Technology (KAIST) and one of the most prominent researchers in systems and cybersecurity. His research spans mobile and wireless security, authentication systems, privacy, and software security, and he has made influential contributions to the security of smartphones, IoT devices, and wireless networks. He is an IEEE Fellow and has served in numerous leadership roles within the security research community.
- **Dr. Monisha Ghosh (University of Notre Dame).** Dr. Monisha Ghosh is a Professor of Electrical Engineering at the University of Notre Dame and a leading expert in wireless communications, spectrum policy, and dynamic spectrum sharing. She previously served as Chief Technology Officer of

the Federal Communications Commission (FCC) and has held leadership roles at the National Science Foundation, Bell Labs, Philips Research, and InterDigital. She is an IEEE Fellow recognized for contributions to cognitive radio and communication systems.

- **Dr. Thyaga Nandagopal (Samsung Research America).** Dr. Thyaga Nandagopal is Head of Emerging Technologies at Samsung Research America, where he leads efforts in 6G, artificial intelligence, robotics, and other future technologies. Prior to Samsung, he held senior leadership positions at the National Science Foundation, including Division Director for Technology, Innovation, and Partnerships, where he helped shape major national research initiatives in networking, AI, semiconductors, and advanced communications.
- **Dr. Edward Oughton (George Mason University).** Dr. Edward Oughton is an Assistant Professor in Geography and Geoinformation Sciences at George Mason University and a leading researcher on telecommunications policy, techno-economics, infrastructure assessment and risk analysis. His work focuses on modeling and analyzing broadband, wireless, satellite, and critical communications networks, having advised more than 20 governments on telecommunications policy, as well as international organizations, and industry stakeholders.
- **Dr. David Love (Purdue University).** Dr. David Love is the Nick Trbovich Professor of Electrical and Computer Engineering at Purdue University and one of the leading researchers in wireless communications. His work has made foundational contributions to MIMO systems, wireless information theory, beamforming, machine learning for communications, and 5G/6G technologies. He is an IEEE Fellow and recipient of numerous IEEE best paper and career awards.

Advisory Committee

The advisory committee brings together internationally recognized experts from academia, industry, and standards organizations with extensive experience in wireless communications, network security, spectrum policy, and next-generation network architectures. The committee will provide strategic guidance on the workshop scope, technical themes, speaker selection, and the formulation of a research roadmap for secure, resilient, and trustworthy future communication systems. Through their combined expertise, the committee will help ensure that the workshop addresses both fundamental scientific challenges and emerging operational threats, while identifying high-impact research directions that can inform future investments, standards development, and technology deployment.

The tentative list of the advisory board is as follows.

- **Greg Pelton (Iridium Communications), Advisory Board Chair.** Mr. Greg Pelton is a senior technology leader at Iridium Communications, where he contributes to the development and deployment of global satellite communications systems. His work focuses on NTN, satellite communications architecture, and the integration of space-based connectivity into next-generation communication ecosystems.
- **Salim Yaghmour (SES).** Mr. Salim Yaghmour is a technology executive at one of the largest satellite operators, SES, with extensive experience in satellite communications, network architecture, and telecommunications strategy. He has played a key role in advancing SES's multi-orbit connectivity solutions and the integration of satellite systems with terrestrial communication networks.
- **Gavin Horn (Qualcomm).** Dr. Gavin Horn is Senior Director of Wireless Research at Qualcomm, where he leads research on next-generation wireless technologies, including 5G-Advanced and 6G

systems. His work spans wireless network architectures, spectrum utilization, artificial intelligence for communications, and emerging mobile technologies.

- **Jeff Reed (Virginia Tech).** Dr. Jeff Reed is the Willis G. Worcester Professor of Electrical and Computer Engineering at Virginia Tech and founder of the Wireless@Virginia Tech research group. He is a leading expert in wireless communications, software-defined radio, spectrum sharing, and dynamic spectrum access, and has made significant contributions to both research and technology commercialization.
- **Jaydee Griffith (ATIS Next G Alliance).** Mr. Jaydee Griffith serves as Managing Director of the ATIS Next G Alliance, a leading industry initiative focused on advancing next-generation wireless technologies and strengthening North American leadership in 6G. He has extensive experience bringing together stakeholders from industry, academia, government, and standards organizations to develop strategic research agendas, technology roadmaps, and collaborative initiatives that accelerate innovation in wireless communications.
- **Mallik Tatipamula (Ericsson).** Dr. Mallik Tatipamula is Chief Technology Officer for Group Function Technologies & Architecture at Ericsson. He is an internationally recognized leader in mobile network evolution, 5G and 6G technologies, cloud-native networks, and telecommunications innovation, and has been actively involved in shaping the future direction of global wireless systems.
- **David Goldman (Starlink).** Mr. David Goldman serves as Vice President of Satellite Policy at Starlink (SpaceX), where he leads regulatory, spectrum, and public policy initiatives related to satellite broadband services. He works closely with government agencies, regulators, and international organizations on issues related to NTN, spectrum management, and global satellite connectivity.
- **Bart Slowik (SylLab Systems).** Mr. Bart Slowik serves as the Chief Executive Officer and Founder of the venture-backed startup SylLab Systems which helps organizations secure NTN systems for the post-quantum world. Mr. Slowik brings expertise in quantum-resistant cryptography, AI-enabled edge systems and resilient space infrastructure.

Location and Time

The workshop will be held over one and a half days, in a January-February timeframe at GMU's FUSE building in Arlington, VA. Prior to the workshop, the organizers will convene a closed-door session with selected industry stakeholders to identify and prioritize potential threat scenarios, such as intelligent jamming, rogue network elements, feeder-link disruptions, and cross-domain cascading failures spanning space, air, and ground segments. These operationally grounded scenarios will serve as inputs to the workshop breakout sessions, ensuring that the resulting research roadmap addresses practical security challenges faced by current and emerging NTN deployments rather than focusing solely on theoretical vulnerabilities. This engagement is particularly important given the growing convergence of civilian NTN systems and threat models traditionally associated with electronic warfare, where advanced disruption, deception, and denial capabilities are increasingly relevant to commercial and critical-infrastructure communications networks.

Selection and Recruitment of Attendees

Workshop participants will be selected through two complementary mechanisms. First, the organizing committee will extend invitations to leading researchers and practitioners from academia, industry, government,

and standards organizations whose expertise aligns closely with the workshop themes. This invitation-based process will help ensure broad representation across disciplines, sectors, technical perspectives, and career stages, while engaging individuals with demonstrated leadership in areas central to the workshop. Second, a portion of the participants will be selected through an open application process. Applicants will be invited to submit a one-page white paper describing a research challenge, emerging threat, innovative solution, or future research direction relevant to the workshop.

Submissions will be evaluated based on their relevance to the workshop objectives, the significance and originality of the proposed ideas, technical merit, and the diversity of perspectives they contribute. Particular consideration will be given to submissions that introduce novel problem formulations, interdisciplinary approaches, innovative methodologies, or new evaluation frameworks. To maximize engagement and collaboration, preference will be given to applicants who commit to participating in the full workshop program, including all technical sessions, breakout discussions, and roadmap-development activities.

Important Dates

The tentative schedule for the workshop is as follows.

- Submission deadline: Sep 15, 2026
- Acceptance notification: Oct 15, 2026
- Registration deadline: Dec 15, 2026
- Workshop dates: Jan. 2027

Program Details

The workshop will be organized over one and a half days and will combine invited presentations, panel discussions, and interactive breakout sessions designed to foster collaboration across academia, industry, government, and standards organizations. The program will feature three keynote presentations delivered by leading experts representing these stakeholder communities, providing perspectives on emerging technological trends, operational challenges, policy considerations, and future research opportunities related to the security and resilience of NTN.

To encourage cross-disciplinary dialogue, the workshop will include four thematic panel discussions focused on critical challenges facing future integrated communication infrastructures. Potential topics include NTN security architectures, resilience against electronic warfare and jamming, AI-native communications and networking, and the secure integration of terrestrial and satellite systems. Each panel will be followed by moderated breakout sessions that enable participants to engage in focused technical discussions and identify key gaps in current research and practice. Moderators will be research, industry or government leaders with sufficient authority to lead discussions, supported by hand-selected early career researchers.

The breakout session topics will be refined based on input gathered from participants before and during the workshop to ensure that discussions reflect the most pressing challenges facing the community. Each breakout group will be tasked with identifying high-impact research problems, barriers to deployment, opportunities for collaboration, and priorities for future investment. The findings from the breakout sessions will be synthesized during a concluding plenary discussion. Finally, a community-driven research roadmap

will be developed that outlines the scientific, technical, and policy challenges to be addressed for making secure and resilient NTN systems a reality.

Family Care Resources

The organizers are committed to fostering an accessible environment that enables participation by researchers and practitioners with family care responsibilities. Prior to the workshop, the organizers will compile and distribute information on available childcare and family care resources in the Ballston/Virginia Square area. This information will include licensed childcare providers, temporary childcare services, family-friendly accommodations, and other relevant local resources to help participants make appropriate care arrangements during their travel.

The workshop website will include a dedicated section that describes these resources, along with links to relevant service providers and local information. Participants will also be invited to identify family care needs during registration so that the organizers can provide additional guidance and facilitate access to available resources whenever possible.

By proactively identifying and communicating family care options, the workshop aims to reduce barriers to participation and enable attendance by individuals whose ability to participate may depend on the availability of such resources. These efforts are intended to support broad participation by researchers and practitioners at all career stages and with heterogeneous personal and family circumstances.

Broader Impacts

The security and resilience of NTNs are critical to a broad range of stakeholders, including government agencies, commercial satellite operators, telecommunications providers, critical infrastructure operators, defense organizations, and the general public. As NTNs increasingly provide global connectivity, PNT services, and support essential sectors such as energy, transportation, and emergency response, vulnerabilities in these systems could have far-reaching economic, societal and national security consequences. This workshop will provide a unique forum to collaborate on associated security challenges facing current and future NTN deployments. The resulting research agenda is expected to inform federal agencies, industry stakeholders, the defense industry, and standards organizations as they develop strategies, architectures, and policies for securing future space-based communications infrastructure. In the longer term, the workshop will foster new interdisciplinary research collaborations, raise awareness of emerging security threats to NTNs, and help establish a community of researchers equipped to address the scientific and technological challenges of securing future space, air, and ground communication systems.

Workshop Report and Dissemination

A primary outcome of the workshop will be a community-driven report summarizing the key findings, research challenges, and both technical and policy recommendations identified during the workshop. The report will synthesize insights from keynote presentations, panel discussions, breakout sessions, and participant feedback to provide a comprehensive assessment of the security and resilience challenges facing future NTNs.

Moreover, the final written output will identify critical research gaps, emerging threats, opportunities for interdisciplinary collaboration, and priorities for future investment by the research community, indus-

try, standards organizations, and government agencies. Particular emphasis will be placed on challenges associated with integrating terrestrial and non-terrestrial networks, electronic warfare and jamming threats, AI-native communication systems, spectrum sharing, network resilience, and the security implications of future 6G architectures.

Following the workshop, the organizers will prepare a draft report and circulate it among participants and advisory committee members for feedback. The final report will be made publicly available through the workshop website and disseminated to relevant research communities, standards organizations, government stakeholders, and industry partners. The organizers will also explore publication opportunities through professional societies and community forums to maximize the visibility and impact of the workshop findings.

In addition to documenting the state of the art, the report will serve as a roadmap for future research by identifying high priority technical challenges, potential research thrusts, recommended testbed and dataset needs, and opportunities for collaboration across academia, industry, and government. The report is expected to provide a valuable resource for researchers, funding agencies, standards bodies, and practitioners working to advance secure, resilient, and trustworthy next-generation communication systems.

Results from prior NSF support

PI: Edward Oughton (GMU). 2434136; \$84,999; 2024–2026. *Collecting Perishable Critical Infrastructure Operational Data for May 2024 Space Weather Events*. **Intellectual Merit:** Collected perishable operational data from critical infrastructure operators and modeled impacts during solar maximum, with a focus on the satellite, power and aviation sectors. **Broader Impacts:** Supported internships for more than 10 aspiring scientists. **Publications Listing:** 1 published journal paper [9], 4 papers in peer review [10–13], and 7 conference presentations. **Evidence of Research Products and their Availability:** All papers are on arXiv with a clear statement listing the affiliated open-source GitHub code repository, as well as the Zenodo repository containing all collected and processed datasets. **Renewed Support:** Project still ongoing with a separate collaborative research proposal under review at the same NSF program.

Co-PI: Alex Sprintson (GMU). 2327510, \$210,000, 2023–2025. *CIF: Small: Maximizing Coding Gain in Coded Computing*. **Intellectual Merit:** The project advanced the theoretical foundations and practical deployment of distributed coded computing by developing new frameworks for managing redundancy, mitigating straggling servers, and improving the efficiency, security, and privacy of large-scale distributed computation. **Broader Impacts:** The methodologies developed in the course of this project have broad applicability to cloud computing, edge computing, distributed artificial intelligence, privacy-preserving machine learning, and large-scale data analytics. The project contributed significantly to workforce development by supporting and mentoring Ph.D. students in advanced topics spanning distributed computing, coding theory, machine learning, privacy, and information theory. **Publications Listing:** The project produced several journal and conference publications [14–22]. **Evidence of Research Products and their Availability:** All of the publications are available on IEEE Xplore and ArXiv. **Renewed Support:** N/A.

AI Transparency Usage Statement

AI was used in this proposal for document proofing (checking typos, grammatical and other language issues), as well as image editing. Senior personnel had full oversight of the content and any mistakes remain the purview of the PI and Co-PI.

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